

Introduction

Every measurement that is made in an experiment entails some measurement uncertainty (or standard deviation), which may be small or significant, depending on the measurement method, precision of the measurement instrument, the care or haste with which the measurement is made, skill of the experimentalist, and the number of repeat measurements, if any. The uncertainty may arise from a precision limitation of the measurement instrument, i.e., a reading between the smallest markings on the instrument may have to be estimated, or it may result from an oscillating measurement parameter that does not settle. In any case, each measurement uncertainty will introduce an uncertainty in the result of the experiment.

Why is it important to account for measurement uncertainties in an experiment? The following example illustrates how one may arrive at different conclusions depending on whether the measurement uncertainties are taken into account. The example involves the catalytic hydrolysis of aqueous ethylacetate, $\text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O} \xrightarrow{k} \text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH}$, over a packed bed of catalyst resin in a plug-flow reactor at a known reaction temperature. Aqueous ethylacetate of a known concentration is fed to the reactor, and the conversion of the ethylacetate is determined as function of the volumetric flow rate of the reactor feed. Figure 1 presents the results of four runs at different flow rates.

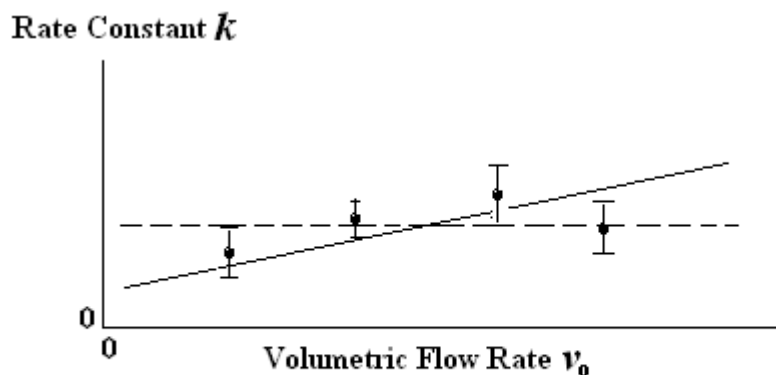


Figure 1. Variation of rate constant with feed flow rate.

If measurement uncertainties are not taken into account, the results would be defined by the four dark circle points alone (i.e., without the error bars), and a linear regression analysis would produce a least-square line that has a positive slope, as shown by the solid line. The latter would lead one to conclude that the reaction rate constant k increases with the volumetric flow rate v_0 and that the reaction is affected by mass transfer.

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On the other hand, if measurement uncertainties are taken into account, the results would then be defined by the data points with the error bars. The error bars indicate that, with the measurement uncertainties, the results are not definitive because all the results within the error bar range are plausible, including a horizontal trend that is marked by the dash line in the figure above. The latter would lead one to conclude that the reaction rate constant is independent of the volumetric feed rate and that the reaction would not be affected by mass transfer. One would have to rely on additional data, such as temperature-variation data, to make a definitive finding. It turns out that additional data support the horizontal trend. Thus, an uncertainties analysis can save us from making an over-reaching conclusion that is embarrassing at least or at worst, costly or disastrous!

At other times, an uncertainties analysis may improve the chances of validating a theory or guide us in determining the correct data trend. Figure 2 presents the experimental results of two series of runs. If measurement uncertainties are not taken into account, one would arrive at the linear regression trends represented by the solid lines, which would indicate different slopes that are contrary to a theory that one is trying to validate. The theory leads one to expect a common slope for the two data series. If measurement uncertainties are taken into account, the data points have error bars that make it possible to draw parallel plots for the two data series, as represented by the dashed lines, thus validating the theory.

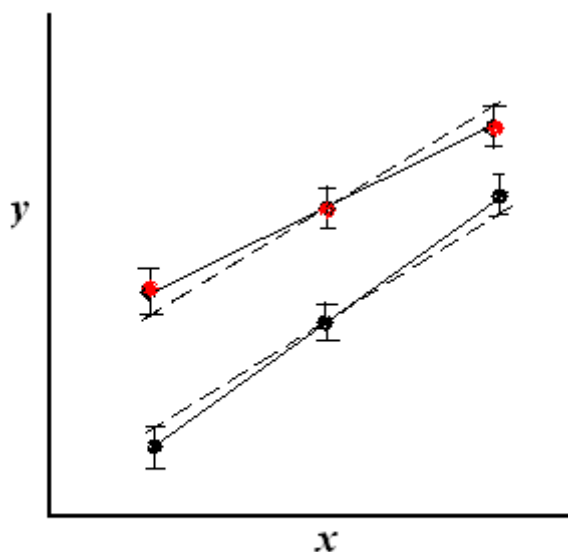


Figure 2. y vs x plots for two series of runs.

The two examples presented above illustrate that experimental results and conclusions could be affected significantly by an uncertainties analysis and that a failure to account for measurement uncertainties could lead to erroneous conclusions.

How Are Measurement Uncertainties Propagated Into an Experimental Result?

Measurement uncertainties are propagated into an experimental result through its functional dependence on the measurement parameters. Let's denote the experimental result or the dependent variable by y and the measurement variables (also known as the independent variables) by x_i , where $i = 1, 2, 3, \dots$. It may be shown that for a general function y that is dependent on independent variables $x_1, x_2, x_3, \dots, x_i$, i.e., $y = f(x_1, x_2, x_3, \dots, x_i)$, the uncertainty

(or standard deviation) of y is related to the uncertainties (or standard deviations) of independent variables $x_1, x_2, x_3, \dots, x_i$ by Eqn (1)

$$\Delta y^2 = \sigma_y^2 = \sum_{i=1}^n \left[\left(\frac{\partial y}{\partial x_i} \right)_{x_{j \neq i}} \right]^2 \sigma_{x_i}^2 = \sum_{i=1}^n \left[\left(\frac{\partial y}{\partial x_i} \right)_{x_{j \neq i}} \right]^2 \Delta x_i^2 \quad (1)$$

Two most common functional dependences of y on x_i are (1) a linear combination relationship, and (2) a power law relationship. It may be shown that, for these two common cases, Equation 1 leads to the following y uncertainty expressions, *provided that the independent variables, x_i ($x_1, x_2, \dots$), are not correlated*. The importance of the last qualification will be illustrated in two later examples.

(1) Linear Combination Relationship

For $y(x_i) = a_0 + a_1 x_1 + a_2 x_2 + \dots$, where $a_0, a_1, a_2, \dots$ are constants that may be positive or negative or zero, and provided that $x_1, x_2, \dots$ are NOT correlated,

$$(\Delta y)^2 = a_1^2 (\Delta x_1)^2 + a_2^2 (\Delta x_2)^2 + a_3^2 (\Delta x_3)^2 + \dots = \sum_{i=1}^n a_i^2 (\Delta x_i)^2 \quad (2)$$

(2) Power Law Relationship

For $y(x_i) = a_0 x_1^{a_1} x_2^{a_2} x_3^{a_3} \dots$, where $a_0, a_1, a_2, \dots$ are constants that may be positive or negative or zero, and provided that $x_1, x_2, \dots$ are NOT correlated,

$$\frac{(\Delta y)^2}{y^2} = a_1^2 \frac{\Delta x_1^2}{x_1^2} + a_2^2 \frac{\Delta x_2^2}{x_2^2} + a_3^2 \frac{\Delta x_3^2}{x_3^2} + \dots = \sum_{i=1}^n a_i^2 \frac{\Delta x_i^2}{x_i^2} \quad (3)$$

Example 1 The unknown density ρ_L of an organic liquid is determined by a buoyancy method. A solid sphere of a known radius R is weighed in the air and then weighed again while submerged in the organic liquid. The following set of data has been obtained: $W_{\text{in air}} = 5.2360 \pm 0.0005$ g, $W_{\text{in liq}} = 1.9478 \pm 0.0005$ g, $R = 1.00 \pm 0.01$ cm. Determine ρ_L and its uncertainty from the data.

Solution

The weight of the solid submerged in the liquid is related to the weight of the solid in air (more rigorously in vacuum) and the buoyancy force B by $W_{\text{in liq}} = W_{\text{in air}} - B$. But the buoyancy force is given by the weight of the liquid displaced, i.e., $B = \frac{4}{3} \pi R^3 \rho_L$.

$$W_{\text{in air}} = \frac{4}{3} \pi R^3 \rho_s \quad \text{where } \rho_s \text{ is the density of the solid.} \quad W_{\text{in liq}} = \frac{4}{3} \pi R^3 (\rho_s - \rho_L)$$

$$\rho_L = \rho_s - \frac{W_{\text{in liq}}}{\frac{4}{3} \pi R^3} = \frac{W_{\text{in air}} - W_{\text{in liq}}}{\frac{4}{3} \pi R^3} = \frac{(5.2360 - 1.9478) \text{ g}}{\frac{4}{3} \pi (1.0)^3 \text{ cm}^3} = 0.7850 \text{ g/cm}^3$$

Using the power law relationship, $\rho_L = \frac{W_{in\ air} - W_{in\ liq}}{\frac{4}{3}\pi R^3} = \frac{(\Delta W)R^{-3}}{\frac{4}{3}\pi}$, it follows from Equation 3

$$\text{that } \frac{\Delta \rho_L^2}{\rho_L^2} = (1)^2 \frac{[\Delta(\Delta W)]^2}{(\Delta W)^2} + (-3)^2 \frac{\Delta R^2}{R^2} = \frac{[\Delta(\Delta W)]^2}{3.2882\ g^2} + 9\left(\frac{0.01}{1.00}\right)^2 \quad (4)$$

but $\Delta W = W_{in\ air} - W_{in\ liq}$, which is a linear combination relationship, it follows from Equation 2 that $[\Delta(\Delta W)]^2 = \Delta W_{in\ air}^2 + \Delta W_{in\ liq}^2 = (0.0005^2 + 0.0005^2)\ g^2$

$$\text{Equation 4 } \Rightarrow \frac{\Delta \rho_L^2}{\rho_L^2} = \frac{[\Delta(\Delta W)]^2}{3.2882^2\ g^2} + 9\left(\frac{0.01}{1.00}\right)^2 = \frac{0.0005^2(2)}{3.2882^2} + 0.0009 = 0.00090005$$

Therefore, $\Delta \rho_L = \pm(0.00090005)^{1/2} \rho_L = \pm(0.0300008)(0.7850)\ g/cm^3 = \pm 0.0236\ g/cm^3$. The density of the organic liquid should then be reported as $\rho_L = (0.7850 \pm 0.0236)\ g/cm^3$.

The above analysis shows that the uncertainty of the density determination is largely dependent on the uncertainty of the radius measurement and the contribution of the uncertainty of mass measurement to the density uncertainty is essentially negligible. If an increased precision of the density determination is desired, the analysis indicates that the incremental effort and resource should be directed at getting a more precise radius measurement rather than mass measurement. In general, an analysis of the propagation of measurement uncertainties is essential for an efficient allocation of resource and effort in seeking an experimental improvement.

Example 2 The following data have been collected on a hot water stream in a heat exchanger experiment: mass flow rate = $5.0 \pm 0.2\ kg/min$, inlet temperature = $60.0 \pm 0.05\ ^\circ C$, outlet temperature = $45.5 \pm 0.05\ ^\circ C$. Given that the specific heat of water is $1.0\ kcal/(kg \cdot ^\circ C)$, determine (a) the heat transfer rate from the hot water stream, and (b) the uncertainty in the heat transfer rate due to measurement uncertainties.

Solution

$$\dot{Q} = -\dot{m}_h \hat{C}_P (T_{h_{out}} - T_{h_{in}}) \equiv -\dot{m}_h \hat{C}_P \Delta T_h \quad (5)$$

$$(a) \ \dot{Q} = -\dot{m}_h \hat{C}_P (T_{h_{out}} - T_{h_{in}}) = -5.0 \frac{kg}{min} * 1.0 \frac{kcal}{kg \cdot ^\circ C} * (45.5 - 60.0)^\circ C = 72.5 \frac{kcal}{min}$$

$$(b) \text{ First Solution that starts with a power law relation, } \dot{Q} = -\dot{m}_h \hat{C}_P \Delta T_h \quad (5)$$

$$\left(\frac{\Delta \dot{Q}}{\dot{Q}} \right)^2 = \left(\frac{\Delta \dot{m}_h}{\dot{m}_h} \right)^2 + \left(\frac{\Delta \Delta T_h}{\Delta T_h} \right)^2 \quad (6)$$

But $\Delta T_h = T_{h_{out}} - T_{h_{in}}$ for which the linear combination rule applies, $\Delta(\Delta T_h)^2 = \Delta T_{h_{out}}^2 + \Delta T_{h_{in}}^2$

$$\left(\frac{\Delta \dot{Q}}{\dot{Q}} \right)^2 = \left(\frac{\Delta \dot{m}_h}{\dot{m}_h} \right)^2 + \left(\frac{\Delta \Delta T_h}{\Delta T_h} \right)^2 = \left(\frac{\Delta \dot{m}_h}{\dot{m}_h} \right)^2 + \frac{(\Delta T_{h_{out}}^2 + \Delta T_{h_{in}}^2)}{(T_{h_{out}} - T_{h_{in}})^2} \quad (6')$$

$$\left(\frac{\Delta\dot{Q}}{\dot{Q}}\right)^2 = \left(\frac{\Delta\dot{m}_h}{\dot{m}_h}\right)^2 + \frac{(\Delta T_{h_{out}}^2 + \Delta T_{h_{in}}^2)}{(T_{h_{out}} - T_{h_{in}})^2} = \left(\frac{0.2}{5.0}\right)^2 + \frac{(0.05^2 + 0.05^2)}{(45.5 - 60.0)^2} = 0.001624$$

$$\Delta\dot{Q} = \pm\dot{Q}\sqrt{0.001624} = \pm 72.5 \frac{\text{kcal}}{\text{min}} * 0.0403 = \pm 2.92 \frac{\text{kcal}}{\text{min}} \quad (7)$$

Second (Erroneous) Solution that starts with the linear combination rule,

$$\dot{Q} = -\hat{C}_P (\dot{m}_h T_{h_{out}} - \dot{m}_h T_{h_{in}}) \quad (5'')$$

would give

$$\Delta\dot{Q}^2 = \hat{C}_P^2 [\Delta(-\dot{m}_h T_{h_{out}})]^2 + \hat{C}_P^2 [\Delta(\dot{m}_h T_{h_{in}})]^2 = \hat{C}_P^2 [\Delta(\dot{m}_h T_{h_{out}})]^2 + \hat{C}_P^2 [\Delta(\dot{m}_h T_{h_{in}})]^2 \quad (8)$$

But $[\Delta(\dot{m}_h T_{h_{out}})]^2$ & $[\Delta(\dot{m}_h T_{h_{in}})]^2$ are given by the power law relation, i.e.,

$$\frac{[\Delta(\dot{m}_h T_{h_{out}})]^2}{(\dot{m}_h T_{h_{out}})^2} = \frac{\Delta\dot{m}^2}{\dot{m}^2} + \frac{\Delta T_{h_{out}}^2}{T_{h_{out}}^2} \Rightarrow [\Delta(\dot{m}_h T_{h_{out}})]^2 = (\dot{m}_h T_{h_{out}})^2 \left[\frac{\Delta\dot{m}^2}{\dot{m}^2} + \frac{\Delta T_{h_{out}}^2}{T_{h_{out}}^2} \right] \text{ or}$$

$$[\Delta(\dot{m}_h T_{h_{out}})]^2 = [T_{h_{out}}^2 \Delta\dot{m}^2 + \dot{m}^2 \Delta T_{h_{out}}^2] \quad (9)$$

$$\frac{[\Delta(\dot{m}_h T_{h_{in}})]^2}{(\dot{m}_h T_{h_{in}})^2} = \frac{\Delta\dot{m}^2}{\dot{m}^2} + \frac{\Delta T_{h_{in}}^2}{T_{h_{in}}^2} \Rightarrow [\Delta(\dot{m}_h T_{h_{in}})]^2 = (\dot{m}_h T_{h_{in}})^2 \left[\frac{\Delta\dot{m}^2}{\dot{m}^2} + \frac{\Delta T_{h_{in}}^2}{T_{h_{in}}^2} \right] \text{ or}$$

$$[\Delta(\dot{m}_h T_{h_{in}})]^2 = [T_{h_{in}}^2 \Delta\dot{m}^2 + \dot{m}^2 \Delta T_{h_{in}}^2] \quad (10)$$

$$\text{Eqn (7), (8) \& (9)} \Rightarrow \Delta\dot{Q}^2 = \hat{C}_P^2 [(T_{h_{out}}^2 + T_{h_{in}}^2) \Delta\dot{m}^2 + \dot{m}^2 (\Delta T_{h_{out}}^2 + \Delta T_{h_{in}}^2)] \quad (11)$$

$$\Delta\dot{Q} = \pm \hat{C}_P \left[(T_{h_{out}}^2 + T_{h_{in}}^2) \Delta\dot{m}^2 + \dot{m}^2 (\Delta T_{h_{out}}^2 + \Delta T_{h_{in}}^2) \right]^{\frac{1}{2}}$$

$$\Delta\dot{Q} = \pm 1.0 \frac{\text{kcal}}{\text{kg} \cdot ^\circ\text{C}} \left[(60^2 + 45.5^2) 0.2^2 + 5.0^2 (0.05^2 + 0.05^2) \right]^{\frac{1}{2}} \left(^\circ\text{C} \frac{\text{kg}}{\text{min}} \right) = \pm 15.06 \frac{\text{kcal}}{\text{min}} \quad (12)$$

The two results given by Equations 7 & 12 are in sharp disagreement. It turns out that the Second Solution is based on an assumption—which turns out to be faulty in this example—that is implicit in the use of the linear combination rule, namely, that the independent variables (xi's) are independent of and not correlated or related to one another. However, in this problem, the two variables upon which the linear combination rule was erroneously applied, $\dot{m}_h T_{h_{out}}$ and $\dot{m}_h T_{h_{in}}$, are, in fact, not independent but are related by $\dot{m}$. Being based on a faulty assumption, the Second Solution is actually not valid.

Example 3 In a study of the head loss coefficient of an expansion valve, the following set of set of data has been collected:

Inlet valve diameter = 0.0225 m, Outlet valve diameter = 0.0296 m

Volumetric flow rate of water through the valve = $(4.10 \pm 0.10) * 10^{-4} \text{ m}^3/\text{s}$

Manometer readings: at valve inlet = $231 \pm 2 \text{ mm H}_2\text{O}$, at valve outlet = $249 \pm 2 \text{ mm H}_2\text{O}$

Given that the mechanical energy balance across the expansion valve is given by

$\frac{P_2 - P_1}{\rho} + \frac{(v_2^2 - v_1^2)}{2} = -K_d \frac{v_1^2}{2}$, where P is the fluid pressure, v is the fluid velocity, ρ is the fluid density ($= 1.0 \text{ g/cm}^3$), K_d is the head loss coefficient, and subscripts 1 and 2 denote inlet and outlet conditions, respectively, determine (a) the head loss coefficient of the expansion valve and (b) its uncertainty.

Solution

$$-K_d = \frac{2(P_2 - P_1)}{\rho v_1^2} + \frac{(v_2^2 - v_1^2)}{v_1^2} = \frac{2(P_2 - P_1)}{\rho v_1^2} + \frac{v_2^2}{v_1^2} - 1 \quad (13)$$

But $v_1 = \frac{4v_o}{\pi d_1^2}$, $v_2 = \frac{4v_o}{\pi d_2^2}$, $P_1 = \rho g h_1$, and $P_2 = \rho g h_2$

$$\therefore -K_d = \frac{2(P_2 - P_1)}{\rho v_1^2} + \frac{v_2^2}{v_1^2} - 1 = \frac{\pi^2 d_1^4 g (h_2 - h_1)}{8 v_o^2} + \frac{d_1^4}{d_2^4} - 1 \quad (13')$$

$$(a) -K_d = \frac{\pi^2 d_1^4 g (h_2 - h_1)}{8 v_o^2} + \frac{d_1^4}{d_2^4} - 1 = \frac{\pi^2 0.0225^4 m^4}{8} 9.81 \frac{m}{s^2} \frac{(0.249 - 0.231)m}{(4.10 * 10^{-4})^2 \frac{m^6}{s^2}} + \left(\frac{0.0225}{0.0296} \right)^4 - 1$$

$$-K_d = 0.3321 + 0.3339 - 1 = -0.3340 \text{ or } K_d = 0.3340$$

(b) Assuming that d_1 and d_2 have no uncertainties, the uncertainty in K_d would then be given by

$$\text{the power law relation, } \left(\frac{\Delta K_d}{K_d} \right)^2 = \left(\frac{\Delta(\Delta h)}{\Delta h} \right)^2 + (-2)^2 \left(\frac{\Delta v_o}{v_o} \right)^2 \quad (14)$$

But $[\Delta(\Delta h)]^2$ is given by the linear combination rule, i.e., $[\Delta(\Delta h)]^2 = (\Delta h_1)^2 + (\Delta h_2)^2$

$$\therefore \left(\frac{\Delta K_d}{K_d} \right)^2 = \left(\frac{(\Delta h_1)^2 + (\Delta h_2)^2}{(h_2 - h_1)^2} \right) + (-2)^2 \left(\frac{\Delta v_o}{v_o} \right)^2 = \frac{(2^2 + 2^2) mm^2}{(249 - 231)^2 mm^2} + 4 \left(\frac{0.10}{4.10} \right)^2 = 0.02707$$

$$\text{or } \Delta K_d = \pm \sqrt{0.02707} K_d = \pm \sqrt{0.02707} * 0.3340 = \pm 0.0550$$

Note that v_1 and v_2 are related by volumetric flow rate $v_o = \pi \left(\frac{d_1}{2} \right)^2 v_1 = \pi \left(\frac{d_2}{2} \right)^2 v_2$

Therefore, they should NOT BOTH be treated as independent variables. If v_1 and v_2 were both treated as independent variables, a different and erroneous result would be obtained (after a somewhat tedious analysis).

Note that while $-K_d = \frac{\pi^2 d_1^4 g (h_2 - h_1)}{8 v_o^2} + \frac{d_1^4}{d_2^4} - 1$ and $-K_d = \frac{\pi^2 d_1^4 g (h_2 - h_1)}{8 v_o * v_o} + \frac{d_1^4}{d_2^4} - 1$ are

mathematically equivalent, the two expressions would yield different ΔK_d results if the power

law relation is applied to the second expression without a consideration of the independency of the parameters. It can be shown that the erroneous approach would yield

$$\left(\frac{\Delta K_d}{K_d}\right)^2 = \left(\frac{\Delta(\Delta h)}{\Delta h}\right)^2 + 2 * (-1)^2 \left(\frac{\Delta v_o}{v_o}\right)^2 = \left(\frac{\Delta(\Delta h)}{\Delta h}\right)^2 + 2 \left(\frac{\Delta v_o}{v_o}\right)^2 \quad (15)$$

$$\left(\frac{\Delta K_d}{K_d}\right)^2 = \frac{(2^2 + 2^2)}{(249 - 231)^2} + 2 \left(\frac{0.10}{4.10}\right)^2 = 0.02588 \quad \text{or} \quad \Delta K_d = \pm \sqrt{0.02588} * 0.3340 = \pm 0.0537,$$

which would be incorrect.

Table 1 summarizes the formulae which are appropriate for the analysis of the propagation of measurement uncertainties for the various functional dependence of the variable y on the independent variables x_i . In addition to the linear combination and power-law relationships covered by Equations 2 and 3 that are most commonly used, Table 1 lists other formulae that are less commonly used.

Table 1. Formulae (derived from Equation 1) for the propagation of measurement uncertainties for the various fundamental dependence of y on x_i .

Function $y = y(x_1, x_2, ..)$	Uncertainty of y , Δy	
$y = a_0 + a_1x_1 + a_2x_2 + \dots$	$\Delta y^2 = a_1^2 x_1^2 + a_2^2 x_2^2 + \dots$	$\Delta y^2 = a_1^2 x_1^2 + a_2^2 x_2^2 + \dots = \sum_{i=1}^n a_i^2 (\Delta x_i)^2$
$y = a_0 + a_1x_1 - a_2x_2 + \dots$	$\Delta y^2 = a_1^2 x_1^2 + (-a_2)^2 x_2^2 + \dots$	
$y = a_o x_1^{a_1} x_2^{a_2} \dots$	$\left(\frac{\Delta y}{y}\right)^2 = a_1^2 \left(\frac{\Delta x_1}{x_1}\right)^2 + a_2^2 \left(\frac{\Delta x_2}{x_2}\right)^2 + \dots$	$\left(\frac{\Delta y}{y}\right)^2 = \sum_{i=1}^n a_i^2 \left(\frac{\Delta x_i}{x_i}\right)^2$
$Z = a_o \frac{x_1^{a_1}}{x_2^{a_2}} x_2^{a_2} \dots = a_o x_1^{a_1} x_2^{-a_2} \dots$	$\left(\frac{\Delta y}{y}\right)^2 = a_1^2 \left(\frac{\Delta x_1}{x_1}\right)^2 + (-a_2)^2 \left(\frac{\Delta x_2}{x_2}\right)^2 + \dots$	
$y = a_o \ln(a_1 * x_1)$ $= a_o \ln a_1 + a_o \ln x_1$	$\Delta y^2 = a_o^2 \left(\frac{\Delta x_1}{x_1}\right)^2$	
$y = a_o \exp(a_1 x_1)$	$\left(\frac{\Delta y}{y}\right)^2 = a_1^2 \Delta x_1^2$	
$y = a_o^{(a_1 x_1)}$	$\left(\frac{\Delta y}{y}\right)^2 = a_1^2 \Delta x_1^2 [\ln(a_o)]^2$	